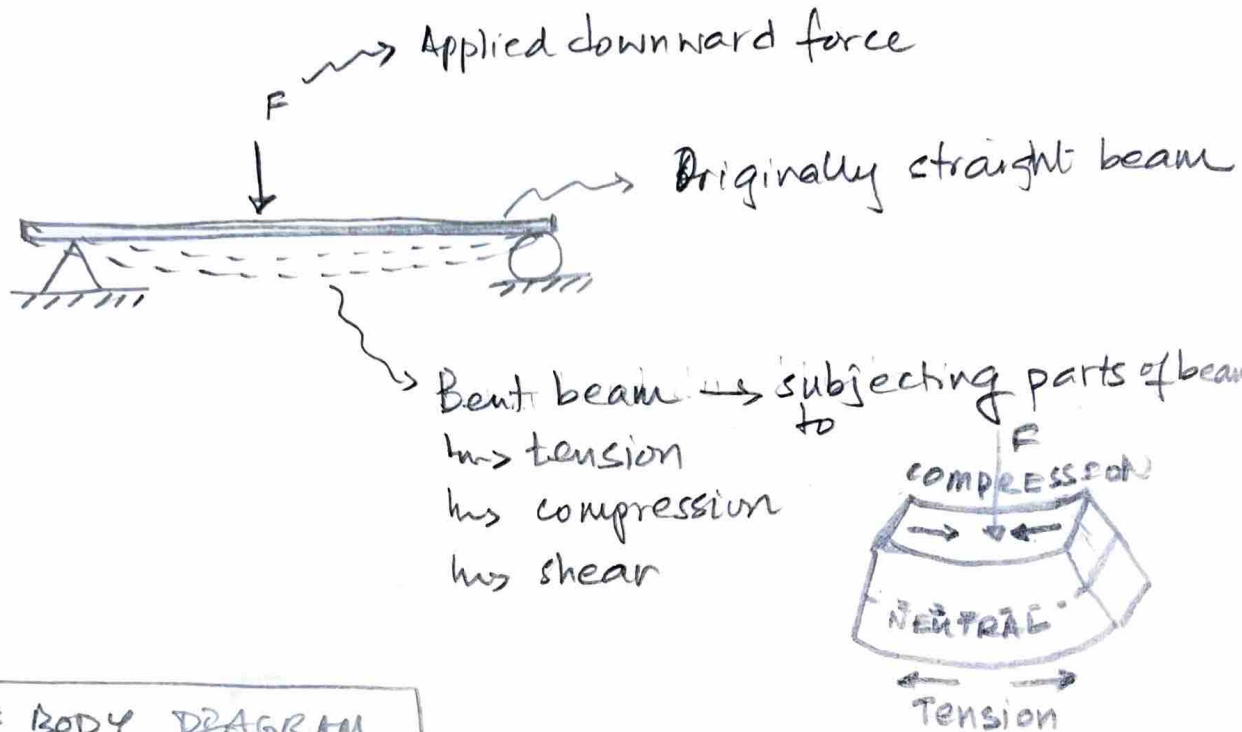
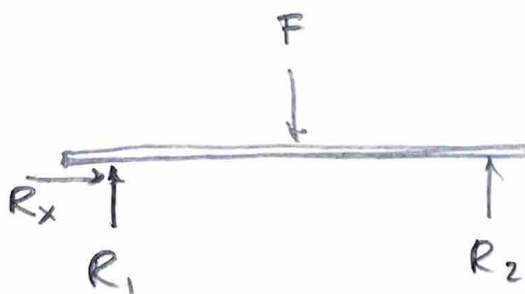


INTRODUCTION TO BENDING:

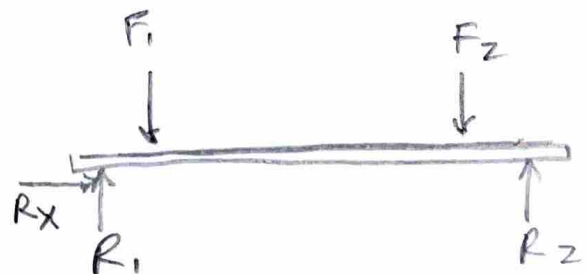
①

CONSIDER A SIMPLY SUPPORTED BEAMFREE BODY DIAGRAM

Three-point bending



Four-point bending



STRESS ANALYSIS
OF
STRUCTURES
SUBJECTED TO
BENDING

STATIC EQUILIBRIUM
ANALYSES

↳ To determine
↳ Internal forces
↳ External forces

DETERMINE STRESSES

②

EQUATIONS FOR 2D PROBLEMS: (X-Y coordinates)

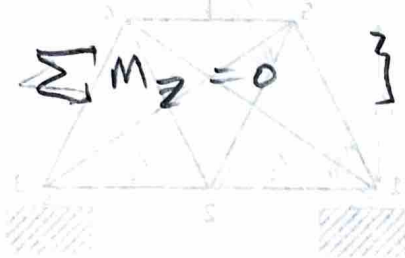
$$\sum F_x = 0$$

$$\sum F_y = 0$$

} TRANSLATIONAL EQUILIBRIUM EQUATIONS

$$\sum M_z = 0$$

} ROTATION EQUILIBRIUM EQUATION

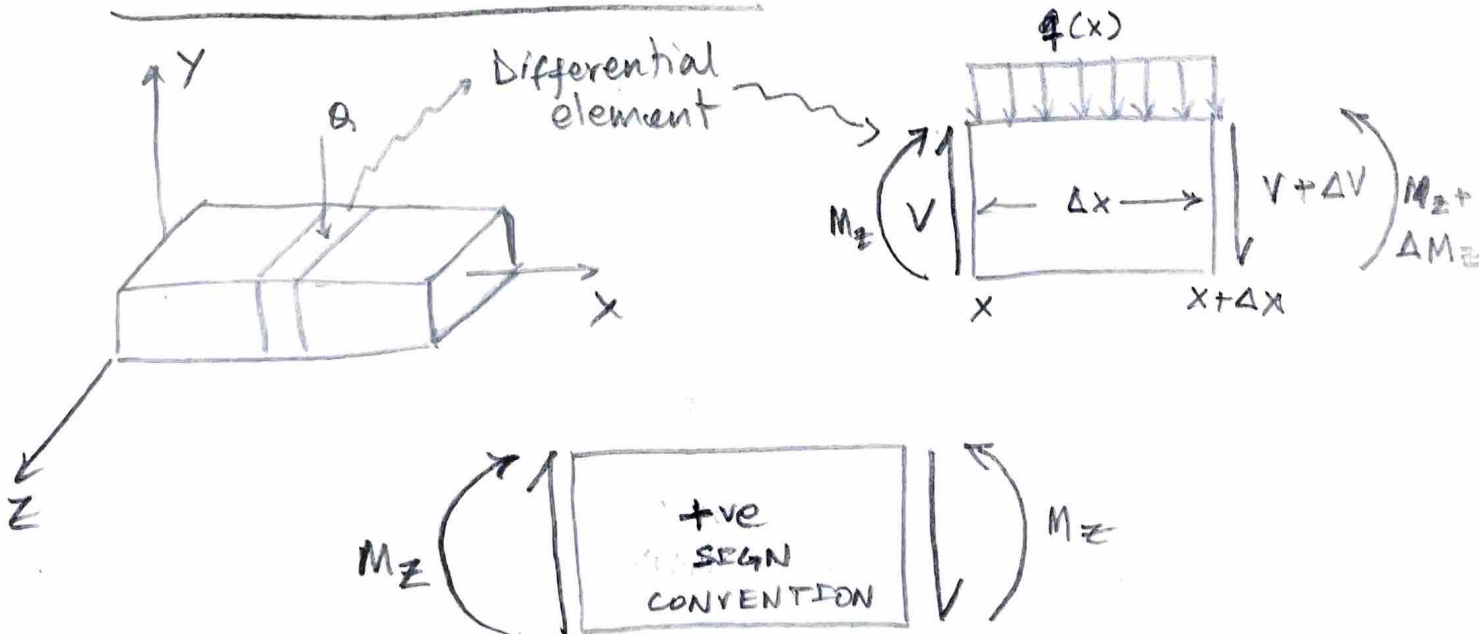


METHOD OF SECTIONS: $\rightarrow$ To determine

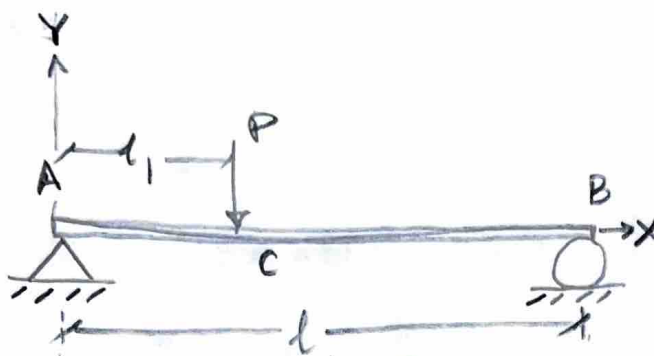
- $\rightarrow$ Internal resistance of structures
- $\rightarrow$ due to externally applied forces

[Individual parts of structures that are in equilibrium]
[should be in equilibrium]

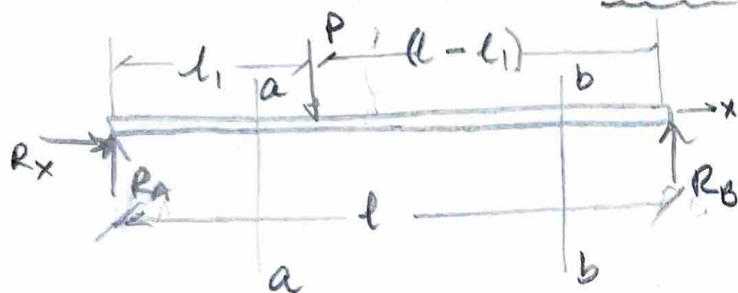
SGN CONVENTION, EQUILIBRIUM ON A DIFFERENTIAL ELEMENT OF A STRUCTURE:



LET'S CONSIDER TWO EXAMPLES:



FREE BODY DIAGRAM OF THE FULL STRUCTURE



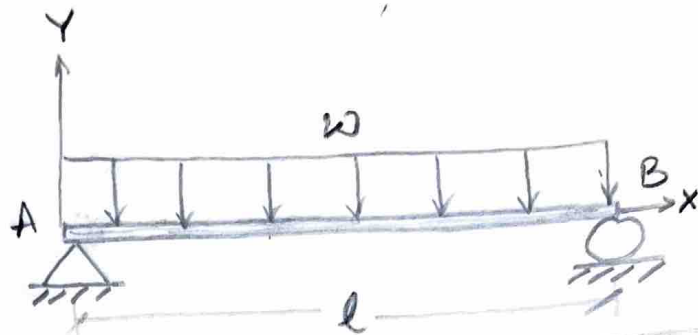
$$\sum F_x = 0 \Rightarrow R_x = 0$$

$$\sum F_y = 0 \Rightarrow R_A + R_B - P = 0$$

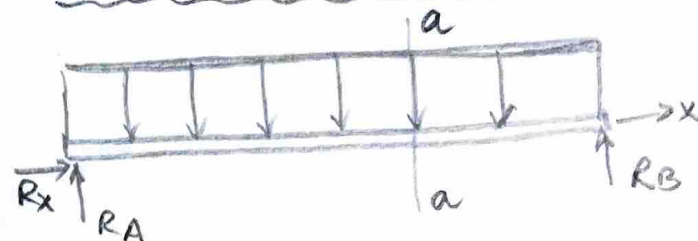
$$\sum M_{EA} = 0 \Rightarrow R_B(l) - P(l_1) = 0$$

$$\Rightarrow R_B = \frac{Pl_1}{l}$$

$$\Rightarrow R_A = P(1 - \frac{l_1}{l})$$



FREE BODY DIAGRAM - FULL STRUCTURE



$$\sum F_x = 0 \Rightarrow R_x = 0$$

$$\sum F_y = 0 \Rightarrow R_A + R_B - \int w dx = 0$$

$$\Rightarrow R_A + R_B = wl$$

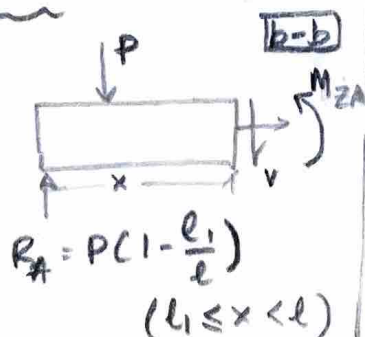
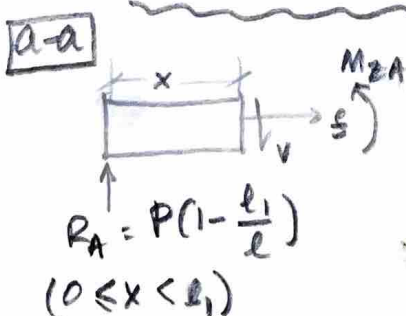
Assume load (wl) is acting at $\frac{l}{2}$ (centroid)

$$\sum M_{ZA} = 0 \Rightarrow R_B l + wl(\frac{l}{2}) = 0$$

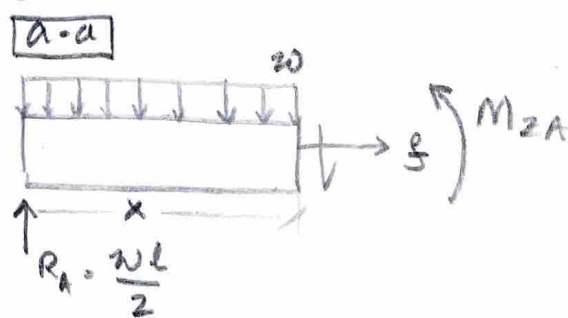
$$\Rightarrow R_B = \frac{wl}{2} \Rightarrow R_A = wl - \frac{wl}{2}$$

$$\Rightarrow R_A = \frac{wl}{2}$$

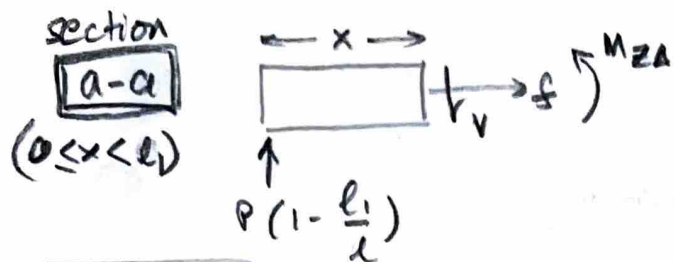
FREE BODY DIAGRAMS - SECTIONS



FREE BODY DIAGRAMS - SECTIONS



WHERE $V \Rightarrow$ SHEAR FORCE, $f \Rightarrow$ FORCE along x-direction



$$\sum F_x = 0 \Rightarrow f = 0$$

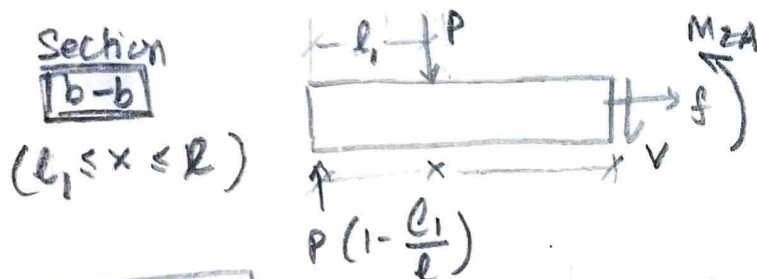
$$\sum F_y = 0 \Rightarrow P(1 - \frac{l_1}{l}) - V = 0$$

$$\Rightarrow V = P(1 - \frac{l_1}{l})$$

$$\sum M_{ZA} = 0 \Rightarrow M_{ZA} - (V)(x) = 0$$

(at section a-a) $\Rightarrow M_{ZA} = (V)(x)$

$$\Rightarrow M_{ZA}(x) = P(1 - \frac{l_1}{l})x$$



$$\sum F_x = 0 \Rightarrow f = 0$$

$$\sum F_y = 0 \Rightarrow P(1 - \frac{l_1}{l}) - P - V = 0$$

$$\Rightarrow V = P - P - P\frac{l_1}{l}$$

$$\Rightarrow V = -P\frac{l_1}{l}$$

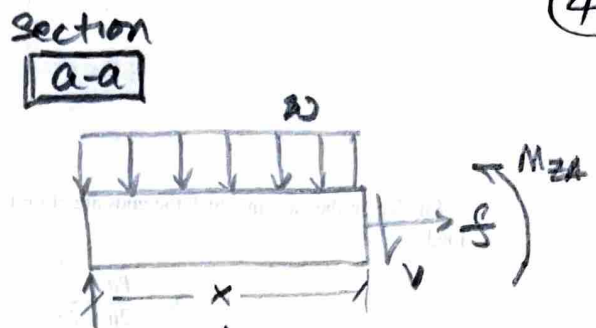
$$\sum M_{ZA} = 0 \Rightarrow M_{ZA} - Pl_1 - (V)(x) = 0$$

$$\Rightarrow M_{ZA} = Pl_1 + (V)(x)$$

$$= Pl_1 + (-\frac{Pl_1}{l})(x)$$

$$\Rightarrow M_{ZA}(x) = Pl_1(1 - \frac{x}{l})$$

(at section b-b)



$$\sum F_x = 0 \Rightarrow f = 0$$

$$\sum F_y = 0 \Rightarrow \frac{wl}{2} - wx - V = 0$$

$$\Rightarrow V = \frac{wl}{2} - wx$$

$$\sum M_{ZA} = 0 \Rightarrow M_{ZA} - (V)(x) - wx(\frac{x}{2}) = 0$$

Load (wx) is acting at $\frac{x}{2}$

$$\Rightarrow M_{ZA} = (V)(x) + (wx)(\frac{x}{2})$$

$$= (\frac{wl}{2} - wx)(x) + (wx)(\frac{x}{2})$$

$$= \frac{wlx}{2} - wx^2 + \frac{wx^2}{2}$$

$$\Rightarrow M_{ZA}(x) = \frac{wlx}{2} - \frac{wx^2}{2}$$

(at section b-b)

This is a quadratic equation (4)

Maximum bending moment occurs when

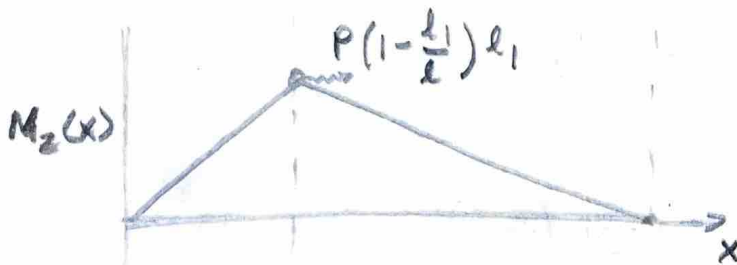
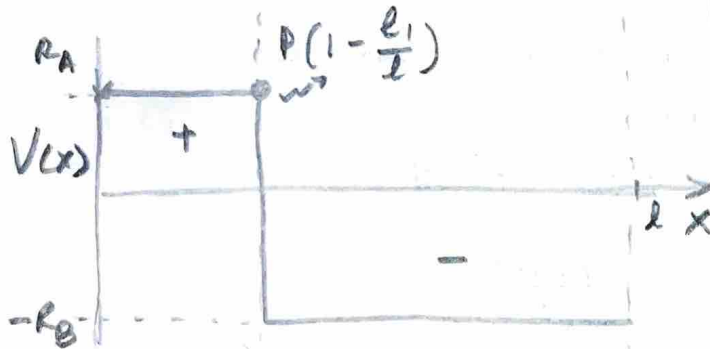
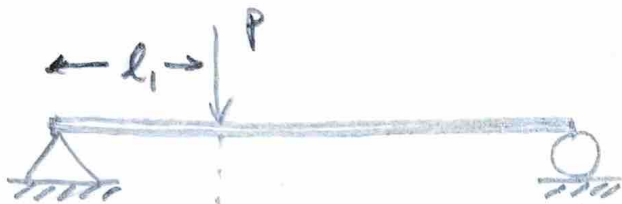
$$\frac{dM_{ZA}(x)}{dx} = 0$$

Shear force results:

$$V(x) = \begin{cases} P(1 - \frac{l_1}{l}) & 0 \leq x < l_1 \\ -P(\frac{l_1}{l}) & l_1 \leq x < l \end{cases}$$

Bending moment results:

$$M(x) = \begin{cases} P(1 - \frac{l_1}{l})x & 0 \leq x < l_1 \\ P l_1 (1 - \frac{x}{l}) & l_1 \leq x < l \end{cases}$$



Maximum bending moment location

$$\frac{dM_{2x}(x)}{dx} = \frac{d}{dx} \left[\frac{wlx}{2} - \frac{wx^2}{2} \right] = 0$$

$$\Rightarrow \frac{wl}{2} - \frac{2wx}{2} = 0$$

$$\Rightarrow \frac{wl}{2} (1 - \frac{x}{l}) = 0 \Rightarrow \boxed{l = \frac{x}{2}}$$

(*) This is the location along x-axis where the bending moment is maximum.

Shear force results:

$$V(x) = \frac{wl}{2} - wx$$

Bending moment results:

$$M(x) = \frac{wlx}{2} - \frac{wx^2}{2}$$

$$M(l/2) = \frac{wl}{2} \frac{l}{2} - \frac{wl^2}{8} = \frac{wl^2}{8}$$

